

Kexin Wei

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Software and robotics engineer with expertise in full-stack development, AI integration, and team leadership. Strong focus on system architecture, quality control, and cross-functional collaboration.

Experience

Research Engineer <i>Imperial Global Singapore.</i>	Mar 2025 — Now <i>Singapore, Singapore</i>
Full-Stack: Built a Flask + Vue.js web application for real-time simulation and cyberattack visualization. Simulation: Translated MATLAB controllers to Python, implemented and tuned a real-time glucose-insulin simulator. DevOps: Dockerized and deployed a closed-loop glucose simulation stack with multi-container orchestration. Mobile: Developed a Flutter based Android app and Vue.js based front end for attendance monitoring. CyberSecurity: Researching agentic automated penetration testing for medical cyber-physical systems. IT: Administers lab workstations, servers, and container infrastructure.	
Software Team Lead <i>Creative Medtech Solutions Pte.Ltd.</i>	Mar 2024 — Jan 2025 <i>Singapore, Singapore</i>
Robot: Engineered robotics control systems including kinematics optimization for 4-DoF robot and control interfaces. CyberSecurity: Enhanced security through encryption protocols, SQL injection protection, and streamlined Qt based GUI. Image: Developed 4D DICOM processing and US/MRI registration algorithms for medical image alignment. Lead: Led Git workflow standardization, Scrum adoption, GoogleTest integration, and code review standards.	
Research and Development Engineer <i>Creative Medtech Solutions Pte.Ltd.</i>	Jun 2022 — Mar 2024 <i>Singapore, Singapore</i>
Robot: Designed motion tracking algorithm for 6 DoF robot arm integrating A* STAR CNN algorithm, verified in animal trial. AI: Developed and validated a AI based 3D medical image segmentation pipeline for prostate biopsy guidance. Image: Prototyped kidney segmentation solutions using SAM, MedSAM, and OpenCV for potential implementation. Sensor: Integrated electromagnetic sensor system in the HIFU treatment software for 3D visualization of the ultrasound probe. Patent: Filed 3 patents in medical algorithm design: HIFU calibration, respiratory tracking, and time synchronization.	

Education

M.Eng in Medical Robotics and AI <i>National University of Singapore, Singapore</i>	Aug 2020 — May 2022 <i>GPA: 4.63/5.00</i>
- Thesis: Deep Reinforcement Learning (DRL) for Robot-assisted Surgical Training - TA for ME2143E Motor Characteristics and ME5405 Machine Vision.	
B.Eng in Mechanical Engineering <i>Tongji University, China</i>	Sep 2014 — Jul 2019 <i>GPA: 4.61/5.00</i>

Awards & Honors

AI Mathematics Modeling Scholarship	Hackathon “Runner Up” Prize at bio LLMs hosted by 4Catalyzer (2023) 2nd Prize in CUMCM (2018); Honorable Mention in MCM/ICM (2018) 3rd Class Award(2018); 1st Class Award(2016); 3rd Class Award(2015)
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Skills

Programming AL/ML CAD/CAE Language	C++, Python, Flask, Vue.js, Linux, Git, flutter, Matlab, SQL, Docker, ROS, Qt, VTK, OpenCV PyTorch, TensorFlow, Deep Learning (CV, CNN, DRL), LLM (Claude, OpenAI, Gemini, LangChain) Solidworks, AutoCAD English (C1), German (C1), Mandarin (native)
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